MAROKA MUSA CALVIN

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PROFESSIONAL SUMMARY

Software Engineering | MACHINE LEARNING DEVELOPER

Motivated Software Engineer and Machine Learning Developer with practical experience in machine learning, data preprocessing, deep learning, and data analysis. Experienced in developing and evaluating AI models using Python and modern machine learning frameworks. Passionate about applying artificial intelligence to solve real-world challenges while continuously expanding technical expertise

TECHNICAL SKILLS

Languages:
Python, , JavaScript, HTML, CSS, C#

Databases:
PostgreSQL, SQL

Backend Development:
Node.js, Flask, Django, REST API

Machine Learning:
PyTorch, Hugging Face, Speech Recognition

Development Tools: Git, GitHub, VS Code, Jupyter Notebook, Kaggle Notebook, Pycharm, .NET Framework, .NET Core, ASP.NET

Data Science: Data Cleaning, Data Preprocessing, Data Analysis, Data Visualization

AI Tools: PyTorch, Hugging Face Transformers

Data Libraries: Pandas, NumPy, Matplotlib, Seaborn, Libros, Scikit-Learn

Systems & Networking:
Linux, Operating Systems, Computer Networking

PROJECTS

Mar 2025 | Cloud Operating System Simulation | 🔗 GitHub

Simulated backend system behaviour for cloud resource management.

- Developed algorithms for dynamic resource allocation
- Implemented concurrency control using synchronization techniques
- Simulated system performance under varying workloads
- Analysed system metrics including CPU and memory usage

Tech: C# (.NET), PostgreSQL

Apr 2026 | Makgobelo Lodge Booking System | Live Link

Developed a full-stack B&B reservation management system with real-world functionality.

• Implemented secure authentication with email verification and role-based access (Admin/User)
• Developed booking system with date validation to prevent double-booking
• Built admin dashboard for managing users, rooms, and booking history
• Integrated email notifications for verification, bookings, and cancellations
• Designed and implemented relational database structure

Tech: Python (Flask), Neon PostgreSQL, HTML, CSS

Feb-Sep 2026 | End-to-End Speech Recognition (ML Project) | GitHub

Developed a machine learning model for speech recognition | Honours Research Project

- Fine-tuned pretrained models using Hugging Face
- Fine-tuned the pretrained XLS-R 300M model.
- Trained on Northern Sotho dataset
- Evaluated model accuracy using Word Error Rate

Tech: Python, PyTorch, Hugging Face

JUN 2026 | MAROKA REAL-TIME MEETING DISCUSSION | Live Link

Developed a real-time meeting discussion platform that supports live communication, saved chat records, user mentions, replies, online user tracking, and meeting recap downloads. NB: Is only used withing an organization

- Implemented real-time messaging using Socket.IO and WebSockets
- Added user join/leave tracking, online user lists, and duplicate-name prevention
- Built saved discussion records for chat history, decisions, tasks, and shared files
- Developed features for mentions, replies, emoji reactions, voice notes, and file sharing
- Prepared the application for deployment using Node.js and Express

Tech: Node.js, Express.js, Socket.IO, JavaScript, HTML, CSS, JSON Storage

CERTIFICATIONS

Python for Machine Learning & Data Science Masterclass — Udemy, 2026 | link

Completed training in Python data science and machine learning, covering real-world datasets, feature engineering, classification, regression, data visualization, NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn.

Python Programming for AWS: Learn Python with AWS and Boto3 — Udemy, 2026 | link

Completed training in using Python with AWS cloud services through Boto3. The course helped me understand how Python can be used to automate cloud tasks, interact with AWS services, and support backend/cloud-based applications.

Python on the Backend — Udemy, 2026 | link
Completed a backend-focused Python course that improved my understanding of server-side development. The course covered how Python is used to build backend systems, process requests, handle application logic, and support web-based applications.

EDUCATION

BSc in Mathematics and Computer Science
University of Limpopo
2023 – 2025

KEY STRENGTHS

- Backend system development and API design
- Knowledge about machine learning frameworks
- Problem solving and system thinking
- Ability to learn and adapt to new technologies quickly
- Experience building real-world software projects